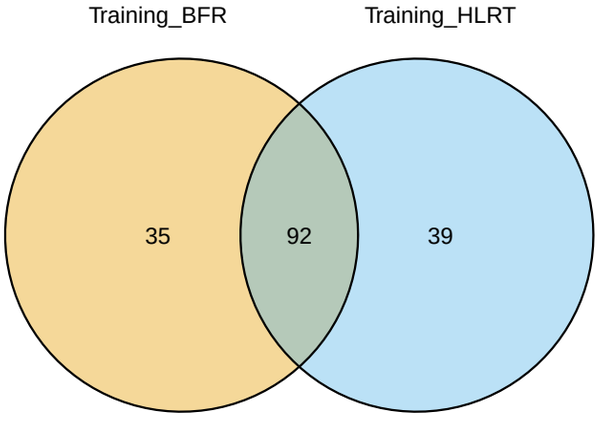
generation of precursor metabolites and energy

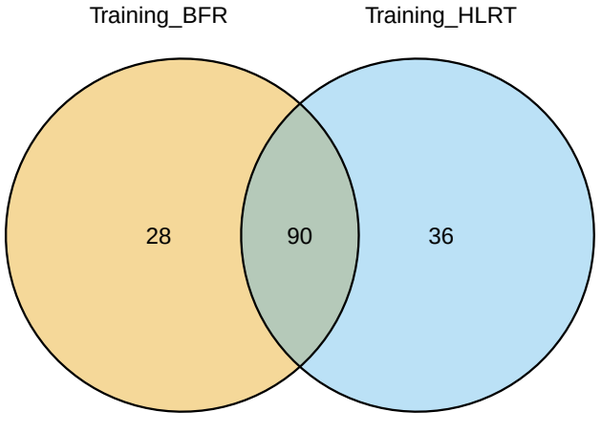
Training_BFR: NES 2.44, padj 1.1e−13
Training_HLRT: NES 2.52, padj 8.9e−17



Shared: ACO2, AKT2, ATP5F1A, ATP5F1C, ATP5F1D, ATP5IF1, ATP5MG, ATP5PB, ATP5PD, ATP5PO, BCKDK, CAVIN3, CHCHD2, CISD1, COQ10A, COX17, COX4I1, COX5B, COX6B1, COX6C, COX7A1, COX7A2, CS, CYC1, DLAT, DLST, ETFA, ETFB, FH, GAA, GBE1, GFPT1, GHITM, GYS1, HCCS, HK1, HMGCL, IDH3A, IDH3B, IDH3G, ISCU, LYRM7, MDH2,...

energy derivation by oxidation of organic compounds

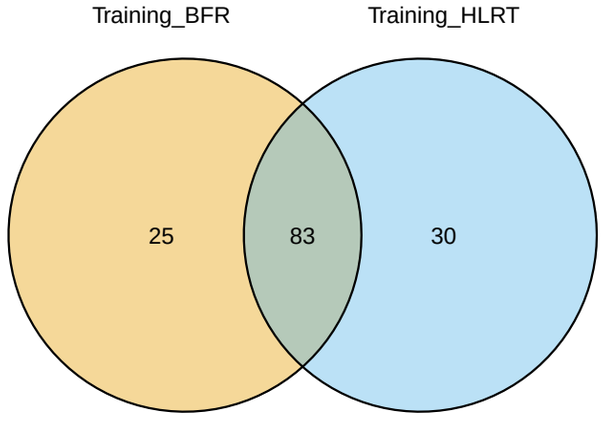
Training_BFR: NES 2.47, padj 1.1e−13
Training_HLRT: NES 2.62, padj 4e−18



Shared: ACO2, AKT2, ATP5F1A, ATP5F1C, ATP5F1D, ATP5IF1, ATP5MG, ATP5PB, ATP5PD, ATP5PO, BCKDK, CAVIN3, CHCHD2, CISD1, COQ10A, COX4I1, COX5B, COX6B1, COX6C, COX7A1, COX7A2, COX7B, CS, CYC1, DLAT, DLST, ETFA, ETFB, FH, GAA, GBE1, GFPT1, GHITM, GYS1, HCCS, HK1, IDH3A, IDH3B, IDH3G, ISCU, LYRM7, MDH2, MT−CO3...

cellular respiration

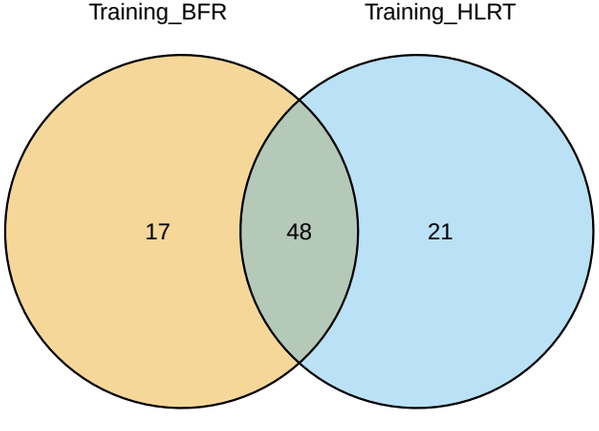
Training_BFR: NES 2.55, padj 2.4e−13
Training_HLRT: NES 2.64, padj 8.9e−17



Shared: ACO2, ATP5F1A, ATP5F1C, ATP5F1D, ATP5IF1, ATP5MG, ATP5PB, ATP5PD, ATP5PO, BCKDK, CHCHD2, CISD1, COQ10A, COX4I1, COX5B, COX6B1, COX6C, COX7A1, COX7A2, COX7B, CS, CYC1, DLAT, DLST, ETFA, ETFB, FH, GHITM, HCCS, HK1, IDH3A, IDH3B, IDH3G, ISCU, LYRM7, MDH2, MT−CO3, MTCH2, NDUFA4, NDUFA9, NDUFB10, NDUF...

mitochondrial gene expression

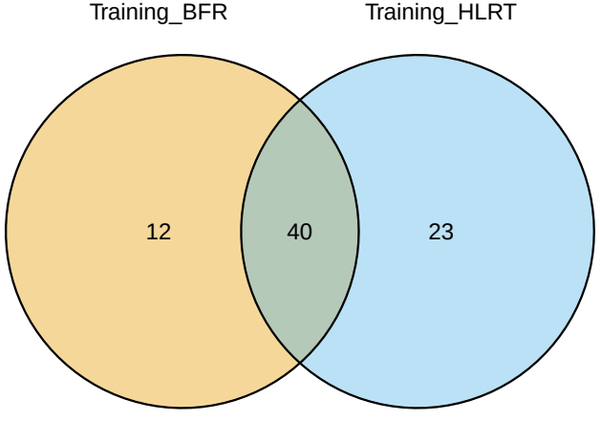
Training_BFR: NES 2.71, padj 1e−11
Training_HLRT: NES 2.84, padj 1.6e−15



Shared: C1QBP, DAP3, DARS2, GFM1, HARS2, HSD17B10, IARS2, LRPPRC, MALSU1, MRPL1, MRPL11, MRPL12, MRPL13, MRPL14, MRPL15, MRPL2, MRPL22, MRPL33, MRPL38, MRPL39, MRPL44, MRPL46, MRPL50, MRPL53, MRPL9, MRPS16, MRPS18B, MRPS23, MRPS24, MRPS27, MRPS28, MRPS30, MRPS34, MRPS35, MRPS9, OXA1L, PNPT1, PRKAA1, PTCD...

mitochondrial translation

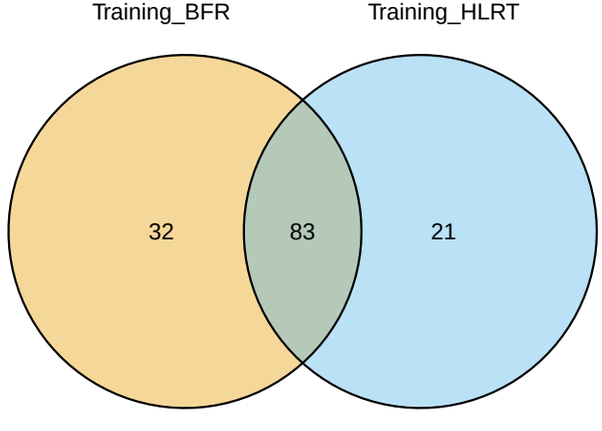
Training_BFR: NES 2.72, padj 1.7e−11
Training_HLRT: NES 2.77, padj 8.6e−14



Shared: C1QBP, DAP3, GFM1, HARS2, IARS2, LARS2, LRPPRC, MRPL1, MRPL10, MRPL11, MRPL12, MRPL13, MRPL14, MRPL15, MRPL2, MRPL22, MRPL33, MRPL38, MRPL39, MRPL44, MRPL46, MRPL50, MRPL53, MRPL9, MRPS16, MRPS18B, MRPS23, MRPS24, MRPS27, MRPS28, MRPS30, MRPS9, OXA1L, PTCD3, SARS2, SHMT2, TACO1, TRMT10C, TSFM, TUFM

aerobic respiration

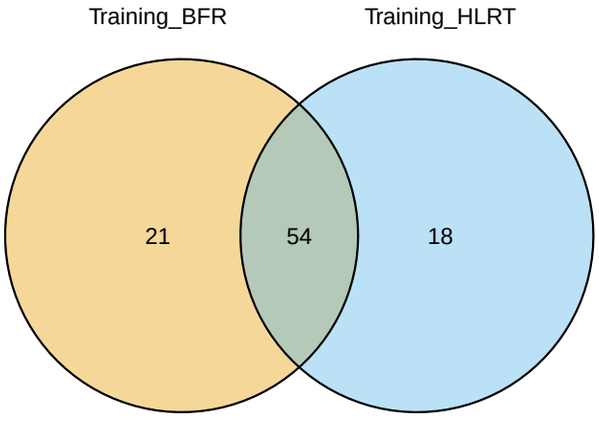
Training_BFR: NES 2.46, padj 2.5e−11
Training_HLRT: NES 2.56, padj 3.8e−14



Shared: ACO2, ATP5F1A, ATP5F1C, ATP5F1D, ATP5IF1, ATP5MG, ATP5PB, ATP5PD, ATP5PO, BCKDK, CHCHD2, COX4I1, COX5B, COX6B1, COX6C, COX7A1, COX7A2, COX7A2L, COX7B, CS, CYC1, DLAT, DLST, FH, GHITM, HK1, IDE, IDH2, IDH3A, IDH3B, IDH3G, ISCU, MDH2, MT−CO3, MTCH2, NDUFA10, NDUFA12, NDUFA4, NDUFA9, NDUFAB1, NDUFB1...

oxidative phosphorylation

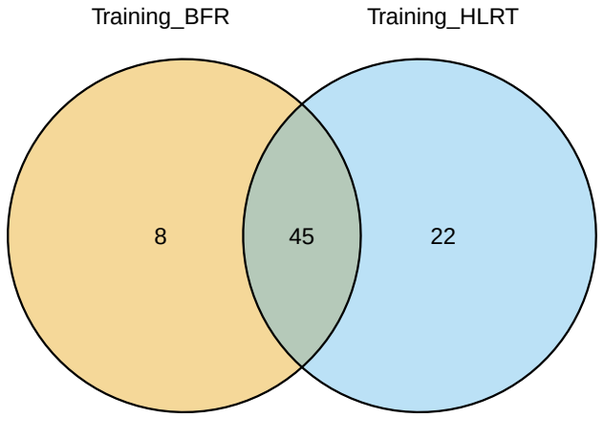
Training_BFR: NES 2.54, padj 7.2e−10
Training_HLRT: NES 2.68, padj 1.5e−13



Shared: ATP5F1A, ATP5F1C, ATP5F1D, ATP5IF1, ATP5MG, ATP5PB, ATP5PD, ATP5PO, CHCHD2, COX4I1, COX5B, COX6B1, COX6C, COX7A1, COX7A2, COX7A2L, COX7B, CYC1, GHITM, ISCU, MT−CO3, MTCH2, NDUFA10, NDUFA12, NDUFA4, NDUFA9, NDUFAB1, NDUFB1, NDUFB10, NDUFB11, NDUFB4, NDUFB5, NDUFB7, NDUFB8, NDUFB9, NDUFC2, NDUFS1, ...

proton transmembrane transport

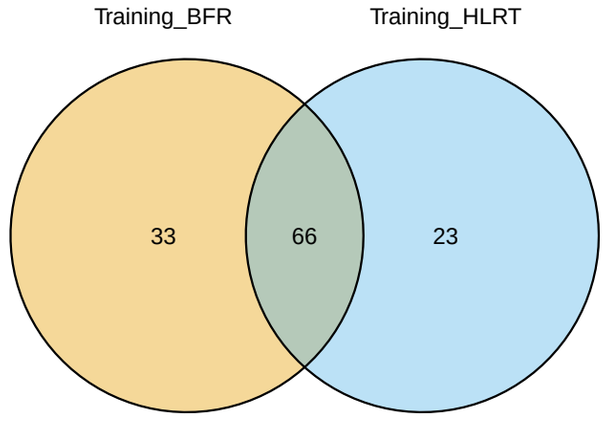
Training_BFR: NES 2.52, padj 2.8e−09
Training_HLRT: NES 2.72, padj 1.5e−13



Shared: ATP1A1, ATP5F1A, ATP5F1C, ATP5F1D, ATP5MG, ATP5PB, ATP5PD, ATP5PO, COX17, COX4I1, COX5B, COX6B1, COX7A1, COX7A2, COX7B, CYC1, GHITM, LAMP1, LETM1, MT−CO3, NDUFA4, NDUFA9, NDUFB10, NDUFB5, NDUFB7, NDUFB8, NDUFB9, NDUFC2, NDUFS1, NDUFS3, NDUFS4, NDUFS6, NDUFS8, NDUFV1, NDUFV2, NNT, PHB2, PPIF, SLC1...

mitochondrion organization

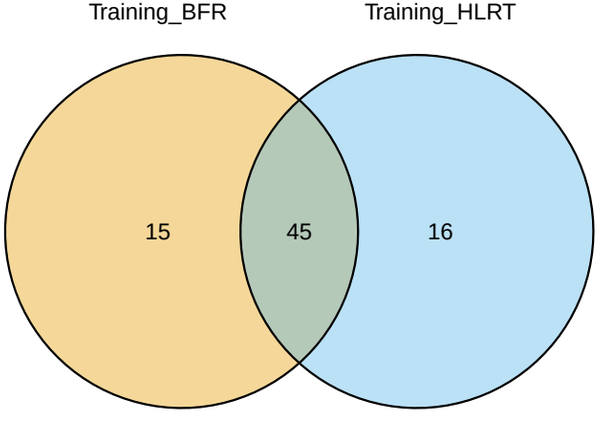
Training_BFR: NES 2.22, padj 2e−08
Training_HLRT: NES 2.24, padj 1.1e−09



Shared: ACAD9, AFG3L2, AIFM1, ATP5F1D, ATP5IF1, ATPAF1, BAX, C12orf73, CHCHD2, CHCHD3, CLPP, COA4, COA6, COX17, COX7A1, DAP3, ENDOG, GHITM, HMGCL, HSD17B10, HSPA9, HSPD1, HTRA2, LETM1, LONP1, LYRM7, MARCKS, MICOS13, MTCH2, MTX1, NDUFAF2, NDUFAF3, NDUFAF4, NDUFC2, NDUFS1, NDUFS3, NDUFS4, NDUFS6, NDUFS8, O...

respiratory electron transport chain

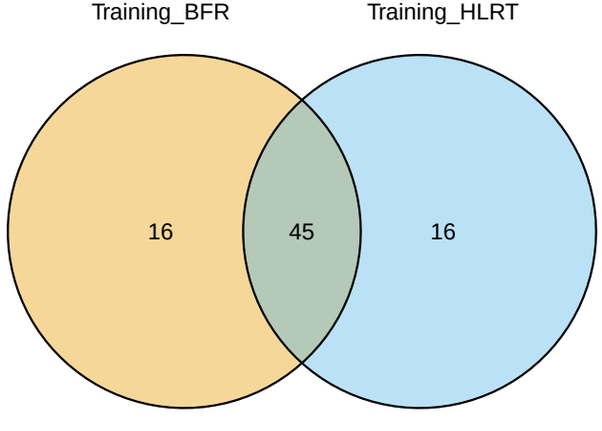
Training_BFR: NES 2.46, padj 4e−08
Training_HLRT: NES 2.71, padj 9e−13



Shared: CHCHD2, COX4I1, COX5B, COX6B1, COX6C, COX7A1, COX7A2, COX7A2L, COX7B, CYC1, ETFA, ETFB, GHITM, HCCS, ISCU, MT−CO3, MTCH2, NDUFA10, NDUFA12, NDUFA4, NDUFA9, NDUFAB1, NDUFB1, NDUFB10, NDUFB4, NDUFB5, NDUFB7, NDUFB8, NDUFB9, NDUFC2, NDUFS1, NDUFS3, NDUFS4, NDUFS5, NDUFS6, NDUFS8, NDUFV1, NDUFV2, SDH...

electron transport chain

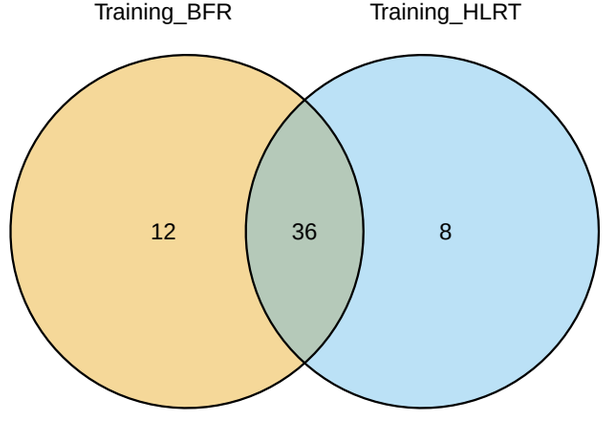
Training_BFR: NES 2.46, padj 1e−07
Training_HLRT: NES 2.64, padj 2.9e−11



Shared: CHCHD2, COX4I1, COX5B, COX6B1, COX6C, COX7A1, COX7A2, COX7A2L, COX7B, CYC1, ETFA, ETFB, GHITM, HCCS, ISCU, MT−CO3, MTCH2, NDUFA10, NDUFA12, NDUFA4, NDUFA9, NDUFAB1, NDUFB1, NDUFB10, NDUFB4, NDUFB5, NDUFB7, NDUFB8, NDUFB9, NDUFC2, NDUFS1, NDUFS3, NDUFS4, NDUFS5, NDUFS6, NDUFS8, NDUFV1, NDUFV2, SDH...

proton motive force−driven mitochondrial ATP synthesis

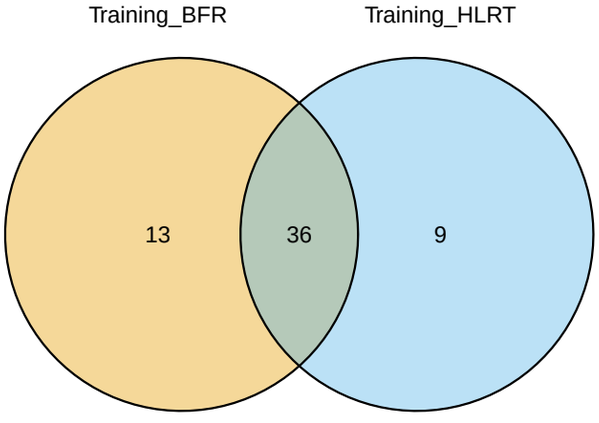
Training_BFR: NES 2.49, padj 2.3e−07
Training_HLRT: NES 2.73, padj 4.9e−11



Shared: ATP5F1A, ATP5F1C, ATP5F1D, ATP5IF1, ATP5MG, ATP5PB, ATP5PD, ATP5PF, ATP5PO, MT−ND1, NDUFA10, NDUFA12, NDUFA8, NDUFA9, NDUFAB1, NDUFB1, NDUFB10, NDUFB11, NDUFB4, NDUFB5, NDUFB6, NDUFB7, NDUFB8, NDUFB9, NDUFC2, NDUFS1, NDUFS3, NDUFS4, NDUFS5, NDUFS6, NDUFS7, NDUFS8, NDUFV1, NDUFV2, SDHA, STOML2

proton motive force−driven ATP synthesis

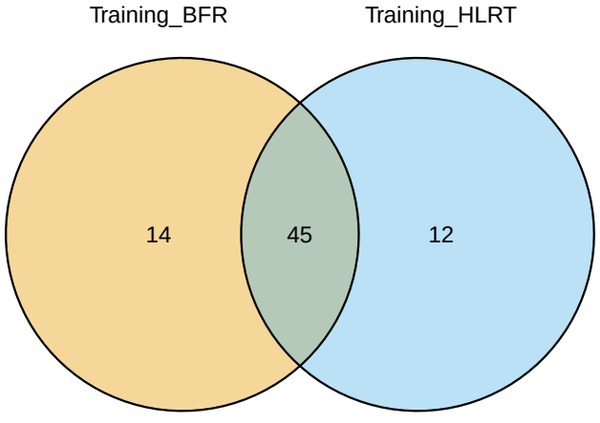
Training_BFR: NES 2.51, padj 4.1e−07
Training_HLRT: NES 2.76, padj 1e−11



Shared: ATP5F1A, ATP5F1C, ATP5F1D, ATP5IF1, ATP5MG, ATP5PB, ATP5PD, ATP5PF, ATP5PO, MT−ND1, NDUFA10, NDUFA12, NDUFA8, NDUFA9, NDUFAB1, NDUFB1, NDUFB10, NDUFB11, NDUFB4, NDUFB5, NDUFB6, NDUFB7, NDUFB8, NDUFB9, NDUFC2, NDUFS1, NDUFS3, NDUFS4, NDUFS5, NDUFS6, NDUFS7, NDUFS8, NDUFV1, NDUFV2, SDHA, STOML2

ATP synthesis coupled electron transport

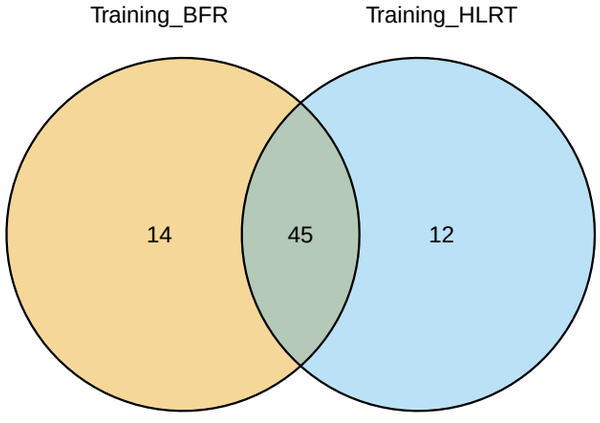
Training_BFR: NES 2.38, padj 7.7e−07
Training_HLRT: NES 2.63, padj 1e−10



Shared: CHCHD2, COX4I1, COX5B, COX6B1, COX6C, COX7A1, COX7A2, COX7A2L, COX7B, CYC1, GHITM, ISCU, MT−CO3, MT−ND1, MTCH2, NDUFA10, NDUFA12, NDUFA4, NDUFA8, NDUFA9, NDUFAB1, NDUFB1, NDUFB10, NDUFB4, NDUFB5, NDUFB6, NDUFB7, NDUFB8, NDUFB9, NDUFC2, NDUFS1, NDUFS3, NDUFS4, NDUFS5, NDUFS6, NDUFS8, NDUFV1, NDUFV...

mitochondrial ATP synthesis coupled electron transport

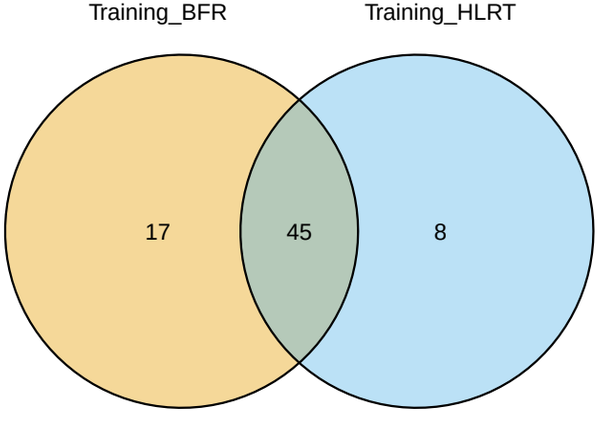
Training_BFR: NES 2.38, padj 7.7e−07
Training_HLRT: NES 2.63, padj 1e−10



Shared: CHCHD2, COX4I1, COX5B, COX6B1, COX6C, COX7A1, COX7A2, COX7A2L, COX7B, CYC1, GHITM, ISCU, MT−CO3, MT−ND1, MTCH2, NDUFA10, NDUFA12, NDUFA4, NDUFA8, NDUFA9, NDUFAB1, NDUFB1, NDUFB10, NDUFB4, NDUFB5, NDUFB6, NDUFB7, NDUFB8, NDUFB9, NDUFC2, NDUFS1, NDUFS3, NDUFS4, NDUFS5, NDUFS6, NDUFS8, NDUFV1, NDUFV...

aerobic electron transport chain

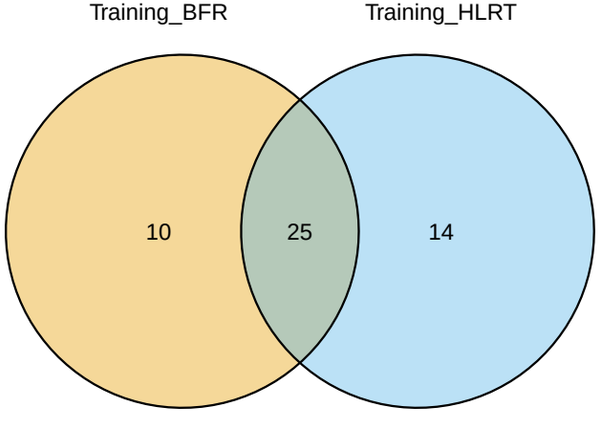
Training_BFR: NES 2.32, padj 1.5e−06
Training_HLRT: NES 2.56, padj 9.7e−10



Shared: COX4I1, COX5B, COX6B1, COX6C, COX7A1, COX7A2, COX7A2L, COX7B, CYC1, ISCU, MT−CO2, MT−CO3, MT−ND1, NDUFA10, NDUFA4, NDUFA8, NDUFA9, NDUFAB1, NDUFB1, NDUFB10, NDUFB4, NDUFB5, NDUFB6, NDUFB7, NDUFB8, NDUFB9, NDUFC2, NDUFS1, NDUFS3, NDUFS4, NDUFS5, NDUFS6, NDUFS7, NDUFS8, NDUFV1, NDUFV2, SDHA, SDHB, ...

mitochondrial respiratory chain complex assembly

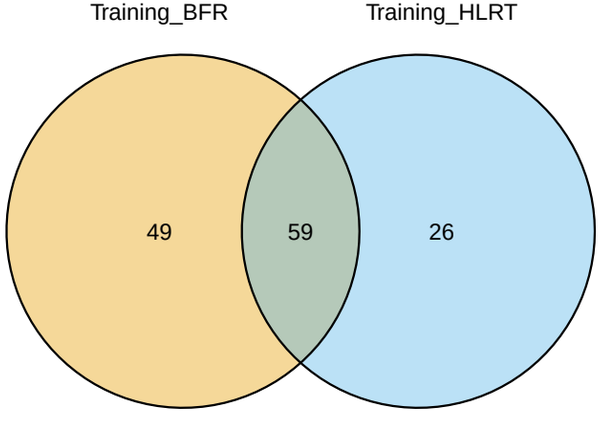
Training_BFR: NES 2.51, padj 1.3e−07
Training_HLRT: NES 2.28, padj 8.9e−06



Shared: ACAD9, AIFM1, ATP5F1D, ATPAF1, C12orf73, CHCHD7, COA4, COA6, COX17, COX20, COX7A1, LYRM7, NDUFAF2, NDUFAF3, NDUFAF4, NDUFS1, NDUFS3, NDUFS4, NDUFS6, NDUFS8, OXA1L, SMIM20, TACO1, TMEM70, UQCRFS1

ubiquitin−dependent protein catabolic process

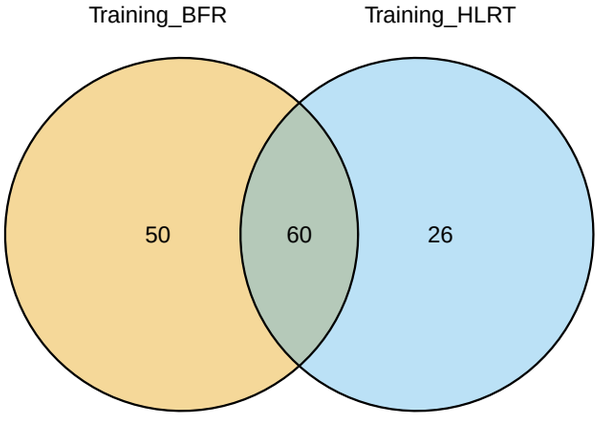
Training_BFR: NES −1.90, padj 3.1e−05
Training_HLRT: NES −2.10, padj 1.1e−07



Shared: ARAF, ARIH1, ARIH2, ASB11, BAG6, CHMP1A, CHMP3, CUL1, CUL3, CUL5, DNAJB2, GIPC1, HECTD1, HUWE1, MTM1, NAA30, NPLOC4, NSFL1C, OTUB1, PARK7, PPP3CA, PRKN, PSMA1, PSMA2, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, PSMC6, PSMD1, PSMD10, PSMD12, PSMD4, RAD23A, R...

modification−dependent macromolecule catabolic process

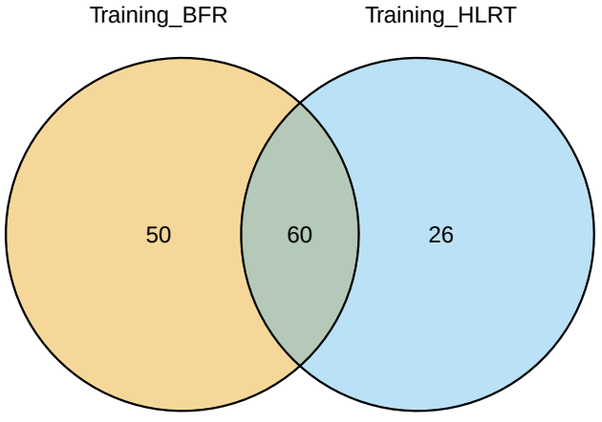
Training_BFR: NES −1.86, padj 9.6e−05
Training_HLRT: NES −2.07, padj 2.4e−07



Shared: ARAF, ARIH1, ARIH2, ASB11, BAG6, CHMP1A, CHMP3, CUL1, CUL3, CUL5, DNAJB2, GIPC1, HECTD1, HUWE1, MTM1, NAA30, NPLOC4, NSFL1C, OTUB1, PARK7, PCYOX1, PPP3CA, PRKN, PSMA1, PSMA2, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, PSMC6, PSMD1, PSMD10, PSMD12, PSMD4, R...

modification−dependent protein catabolic process

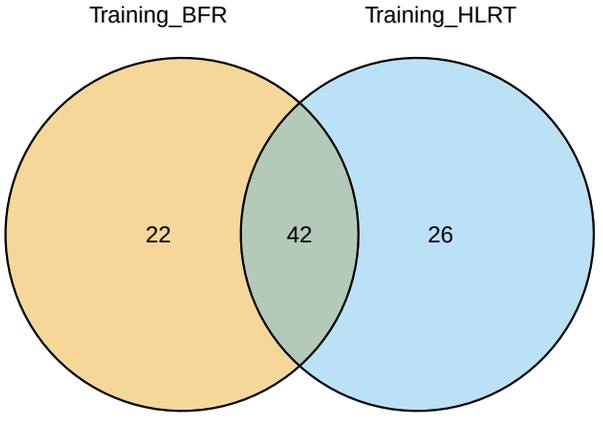
Training_BFR: NES −1.86, padj 0.0001
Training_HLRT: NES −2.08, padj 1.5e−07



Shared: ARAF, ARIH1, ARIH2, ASB11, BAG6, CHMP1A, CHMP3, CUL1, CUL3, CUL5, DNAJB2, GIPC1, HECTD1, HUWE1, MTM1, NAA30, NPLOC4, NSFL1C, OTUB1, PARK7, PCYOX1, PPP3CA, PRKN, PSMA1, PSMA2, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, PSMC6, PSMD1, PSMD10, PSMD12, PSMD4, R...

purine−containing compound biosynthetic process

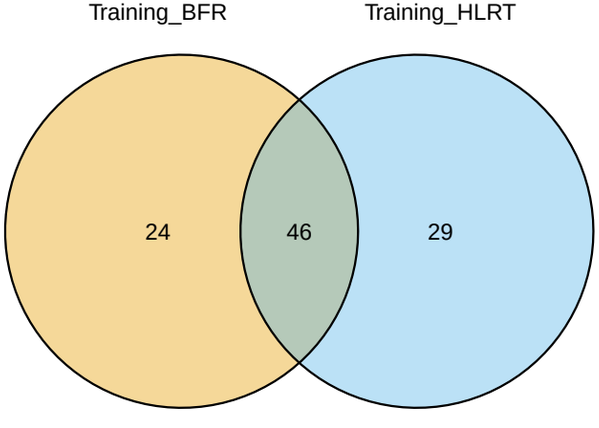
Training_BFR: NES 2.02, padj 9.7e−06
Training_HLRT: NES 1.91, padj 0.00012



Shared: ACSL1, ADSS1, AK2, AK3, APRT, ATP5F1A, ATP5F1C, ATP5F1D, ATP5IF1, ATP5MG, ATP5PB, ATP5PD, ATP5PO, BCKDK, CBR4, DLAT, HSD17B12, NADK2, NAMPT, NDUFA9, NDUFB10, NDUFB11, NDUFB5, NDUFB7, NDUFB9, NDUFC2, NDUFS1, NDUFS3, NDUFS4, NDUFS6, NDUFS8, NDUFV1, NDUFV2, NME1, NNMT, PDHA1, PDK2, PDK4, PDP1, PPCS,...

nucleoside phosphate biosynthetic process

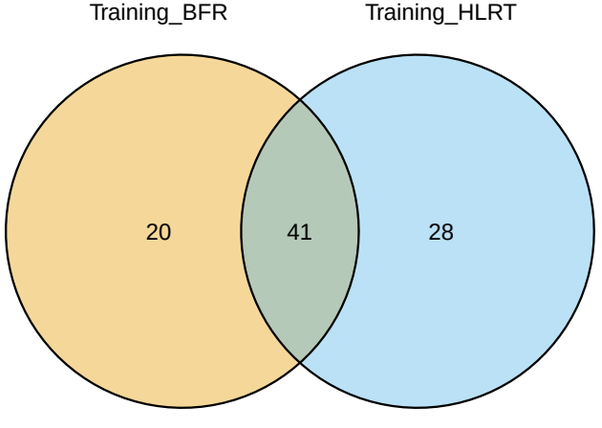
Training_BFR: NES 2.02, padj 6.8e−06
Training_HLRT: NES 1.88, padj 0.00015



Shared: ACSL1, ADSS1, AK2, AK3, APRT, ATP5F1A, ATP5F1C, ATP5F1D, ATP5IF1, ATP5MG, ATP5PB, ATP5PD, ATP5PO, BCKDK, CBR4, DLAT, GFPT1, HK1, HSD17B12, NADK2, NAMPT, NANS, NDUFA9, NDUFB10, NDUFB11, NDUFB5, NDUFB7, NDUFB9, NDUFC2, NDUFS1, NDUFS3, NDUFS4, NDUFS6, NDUFS8, NDUFV1, NDUFV2, NME1, NNMT, PDHA1, PDK2,...

proteasome−mediated ubiquitin−dependent protein catabolic process

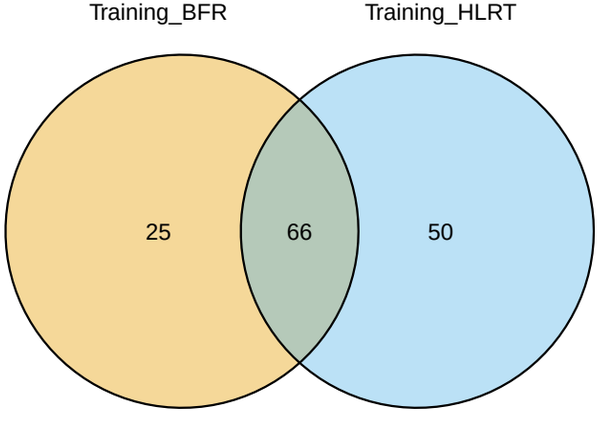
Training_BFR: NES −1.89, padj 0.00016
Training_HLRT: NES −2.14, padj 8.3e−07



Shared: ARIH2, ASB11, BAG6, CUL1, CUL5, DNAJB2, GIPC1, HECTD1, HUWE1, MTM1, NPLOC4, NSFL1C, OTUB1, PARK7, PRKN, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, PSMD1, PSMD10, PSMD4, RAD23A, RNF170, RPS6KB1, STUB1, UBE2B, UBE3A, UBQLN4, UFL1, USP7, YWHAE

post−translational protein modification

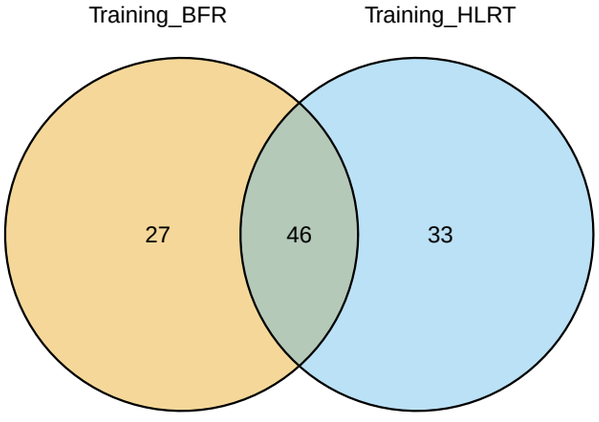
Training_BFR: NES −1.80, padj 0.00017
Training_HLRT: NES −1.84, padj 5e−05



Shared: ARIH1, ARIH2, ASB11, BCL2, BLMH, BRCC3, CAND2, CBFB, CDK5RAP3, CHP1, COPS2, COPS4, COPS5, COPS6, COPS7A, COPS7B, COPS8, CUL1, CUL3, CUL5, DCUN1D2, DNAJB2, GPS1, GSK3B, HECTD1, HINT1, HUWE1, MKRN2, MOCS3, NAE1, NPEPPS, NUB1, OTUB1, PARK7, PLAA, PRKN, PSMD10, RELA, RNF114, RNF126, RNF170, RNF181, S...

proteasomal protein catabolic process

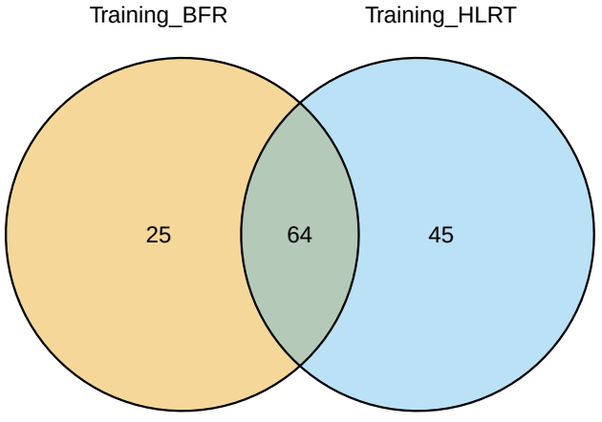
Training_BFR: NES −1.82, padj 0.00025
Training_HLRT: NES −2.05, padj 9.1e−07



Shared: ARIH2, ASB11, BAG6, CUL1, CUL5, DNAJB12, DNAJB2, ECPAS, ERLIN2, GABARAPL2, GIPC1, GPX1, HECTD1, HUWE1, MTM1, NPLOC4, NSFL1C, OTUB1, PARK7, PRKN, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, PSMD1, PSMD10, PSMD4, RAD23A, RNF170, RPS6KB1, STUB1, UBE2B, ...

protein modification by small protein conjugation or removal

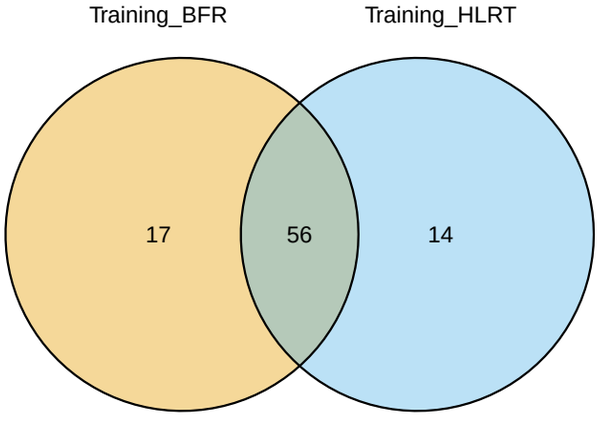
Training_BFR: NES −1.84, padj 0.0001
Training_HLRT: NES −1.82, padj 0.00029



Shared: ARIH1, ARIH2, ASB11, BCL2, BLMH, BRCC3, CAND2, CBFB, CDK5RAP3, CHP1, COPS2, COPS4, COPS5, COPS6, COPS7A, COPS7B, COPS8, CUL1, CUL3, CUL5, DCUN1D2, DNAJB2, GPS1, GSK3B, HECTD1, HINT1, HUWE1, MKRN2, MOCS3, NAE1, NPEPPS, NUB1, OTUB1, PARK7, PLAA, PRKN, PSMD10, RELA, RNF114, RNF126, RNF170, RNF181, S...

muscle contraction

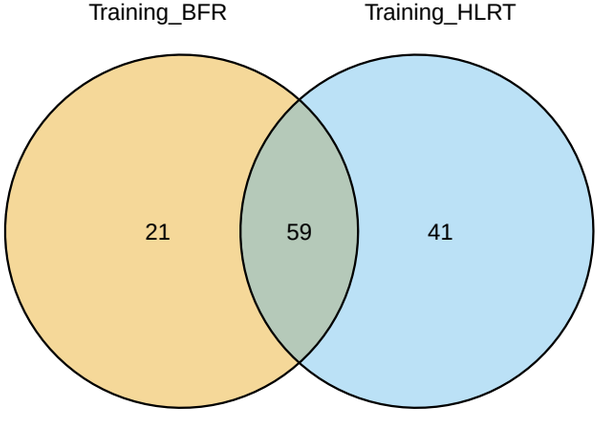
Training_BFR: NES −2.29, padj 4e−08
Training_HLRT: NES −1.88, padj 0.00029



Shared: ABCC9, ACTA1, ACTN3, ATP2A1, ATP2A2, CA3, CACNA1S, CACNA2D1, CACNB1, CACNG1, CASQ1, CD38, CERT1, DMD, DTNA, EMD, ENO1, GSTM2, HRC, JPH1, JSRP1, LMOD1, MAP2K1, MYH1, MYH13, MYH14, MYH7, MYH8, MYL2, MYL3, MYL4, MYL6B, MYLK2, MYLPF, MYOM2, PLN, RCSD1, RYR1, RYR2, SCN4A, SGCA, SGCD, SNTA1, SSPN, STAC...

protein modification by small protein conjugation

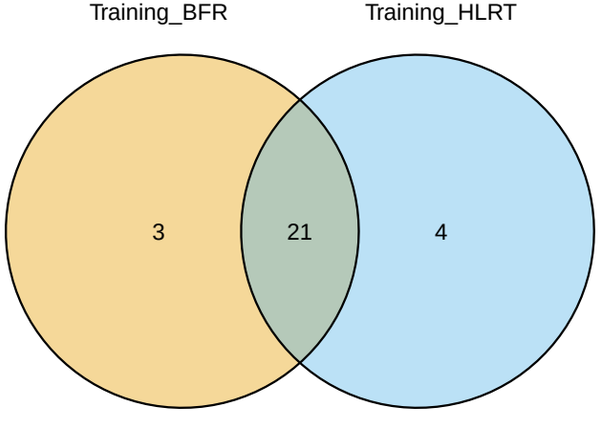
Training_BFR: NES −1.81, padj 0.0003
Training_HLRT: NES −1.85, padj 9.8e−05



Shared: ARIH1, ARIH2, ASB11, BCL2, BLMH, CAND2, CBFB, CDK5RAP3, CHP1, COPS2, COPS4, COPS5, COPS6, COPS7A, COPS7B, COPS8, CUL1, CUL3, CUL5, DCUN1D2, DNAJB2, GPS1, GSK3B, HECTD1, HUWE1, MKRN2, MOCS3, NAE1, NPEPPS, NUB1, PARK7, PLAA, PRKN, PSMD10, RELA, RNF114, RNF126, RNF170, RNF181, SQSTM1, STUB1, THOP1, ...

skeletal muscle contraction

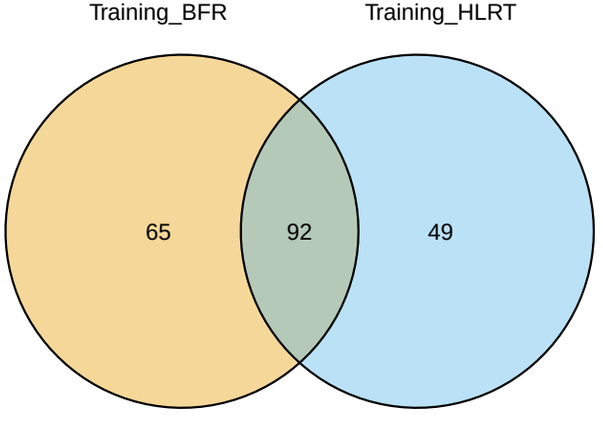
Training_BFR: NES −2.30, padj 6.8e−05
Training_HLRT: NES −2.19, padj 0.00037



Shared: ACTN3, ATP2A1, CA3, CASQ1, DMD, GSTM2, JSRP1, MYH14, MYH7, MYH8, RCSD1, SCN4A, STAC3, SYNM, TCAP, TNNC1, TNNC2, TNNI2, TNNT1, TNNT3, TTN

protein modification process

Training_BFR: NES −1.66, padj 0.00024
Training_HLRT: NES −1.57, padj 0.00042



Shared: ADPRHL2, ARAF, ARIH1, ARIH2, ASB11, BAG6, BCL2, BLMH, CAMK2A, CAND2, CDK5RAP3, CDKN1B, CHMP6, CHP1, COPS2, COPS4, COPS6, COPS7A, COPS8, CTF1, CUL1, CUL3, CUL5, DBI, DEPTOR, DNAJB2, FNTA, GMPPA, GMPPB, GPS1, HECTD1, HHATL, HINT1, HSPB1, HUWE1, INSR, LIMCH1, MACROD1, MAP2K1, MAP2K2, MAP3K20, MOCS3,...